Sabah Nushra

sabahnushra@iut-dhaka.edu

Sabah Nushra | Meowthhh | Meowthhh

Dhaka 1207, Bangladesh

Education

Islamic University of Technology (IUT) *Jan 2020 – Jun 2024*
B.Sc. in Computer Science and Engineering Dhaka, Bangladesh
GPA: 3.25/4.00

Research Interests

Computer Vision, Multimodal & Vision-Language Models, Trustworthy AI, Model Evaluation, Distribution Shift, Uncertainty & Calibration, Test-Time Adaptation, Low-Resource & Multilingual AI

Research Experience

ELITE Research Lab LLC *Jan 2026 – Present*
Research Intern

- Conducted research on trustworthy AI, multimodal and vision-language model evaluation, cross-dataset generalization, uncertainty-aware evaluation, and model reliability across computer vision, medical imaging, and language-model applications.
- Designed reproducible benchmarking pipelines for calibration, selective prediction, distribution shift, dataset leakage and provenance auditing, robustness analysis, and systematic model failure evaluation.

Research Projects

Do Open Language Models Invent Scripture? BILAL-BENCH: Benchmarking Islamic-Source Fabrication in Bengali
Accepted, Muslims in ML Workshop at NeurIPS 2026

- Introduced *BILAL-BENCH*, a 240-question expert-verified Bengali benchmark across five categories, and evaluated seven open-weight models on 1,680 double-annotated responses.
- Found 79.3–95.3% confident fabrication for six of seven models on source-production questions; only 2 of 315 Hadith responses were both correct and correctly sourced, with no consistent scale trend.

Knowing When Not to Answer: A Multi-Domain Benchmark and Lightweight Alignment-Based Abstention for Medical VQA
Under Review, NAACL 2027

- Constructed a balanced benchmark of 45,942 answerable and unanswerable medical VQA pairs across five imaging domains, incorporating visual-grounding filtering and BioMedCLIP-guided negative sampling to minimize exploitable linguistic biases.
- Developed a parameter-efficient abstention framework for frozen LLaVA-Med using question-to-statement reformulation and BioMedCLIP alignment with learned projection layers. Improved correct abstention from 8.5–12.7% to 87.5–93.9% across four imaging domains, while analyzing cross-domain generalization and answerability trade-offs.

CRuSS: Cross-domain Reliability under Shift and Stress for Medical Image Classification
Under Review, Biomedical Signal Processing and Control

- Implemented and evaluated CRuSS, a five-component reliability metric integrating clean predictive performance, cross-domain retention, calibration, corruption robustness, and class-specific resilience across eight CNN, Transformer, and foundation-model architectures on three brain MRI datasets.
- Conducted multi-seed evaluation involving 120 clean runs and 240 cross-dataset tests, alongside controlled corruption analysis and dataset-provenance auditing. Identified model-ranking reversals hidden by conventional accuracy metrics and substantial image overlap that compromised independent external validation.

Understanding How Vision Models Fail: A Graph-Based Framework for Severity, Stability, and Model Complementarity
Manuscript in Preparation

- Developed a graph-based evaluation framework to characterize how vision models fail by capturing the frequency, severity, and structural patterns of class-level prediction errors.
- Developed *Failure Graph Risk (FGR)* from failure frequency, confidence, semantic distance, and sample difficulty, with analyses of stability and model complementarity.

Beyond the Leaderboard: Reliability-Aware Operating-Point Evaluation for Bangla Cyberbullying Moderation
Manuscript in Preparation

- Evaluated Bangla-specific, multilingual, and compact generative models on 43,147 cleaned comments under a frozen train-development-test protocol for five-class cyberbullying moderation.

- Analyzed calibration, selective prediction, rare-class Threat behavior, realistic Bangla perturbations, human errors, and compute cost; the calibrated ensemble achieved Macro-F1 0.8547 and Risk@90 0.0973.

Academic Service

Reviewer, Muslims in ML Workshop at NeurIPS 2026

Selected Projects

Seeing Isn't Structuring *2026*
Python, PyTorch, Hugging Face, Jupyter/Kaggle

- Developed a controlled 170-query multimodal benchmark across seven failure modes, with synthetic Snakes & Ladders generation, prerequisite calibration, automated scoring, and cross-model evaluation of seven open-weight VLMs to identify systematic failures in structured visual reasoning beyond basic perception.

Darkroom: Old Photo Restoration and Colorization *2026*
Python, PyTorch, OpenCV, Gradio, Hugging Face

- Built an end-to-end restoration and colorization pipeline combining denoising, scratch removal, contrast enhancement, and transfer-learned colorization, improving PSNR from 15.28 to 16.84 and SSIM from 0.282 to 0.487 on synthetically degraded test images.

ML Algorithm Visualizer *2024*
Jupyter, scikit-learn

- Developed educational software that visually demonstrates the working processes of core machine-learning algorithms, including Linear and Logistic Regression, Neural Networks, SVMs, K-means, Naive Bayes, Decision Trees, and PCA, to make fundamental concepts easier to understand through visual learning.

Helmet Detection using YOLO *2023*
Python, OpenCV, YOLOv5/YOLOv6/YOLOv8

- Developed a web application to detect whether motorcycle riders are wearing helmets, comparing YOLOv5, YOLOv6, and YOLOv8 on the project dataset for automated road-safety monitoring.

Additional Experience

Beetles Cyber Security Ltd. *Aug 2024 – Dec 2024*
Cybersecurity Engineer Dhaka, Bangladesh

- Assisted with day-to-day security monitoring, alert triage, and vulnerability scanning across systems and network assets, escalating suspicious activity and potential incidents to senior team members.
- Supported basic system and network hardening, remediation follow-up, access-control checks, and security documentation for identified vulnerabilities and incidents.

Leadership

IUT Programming Community *May 2022 – May 2024*
Advisor

- Mentored students in programming, problem solving, data structures, and algorithms; organized national-level contests, workshops, problem-solving sessions, and industry-led software-development seminars.

IEEE IUT Student Branch *Feb 2022 – May 2024*
Chief Public Relations Officer

- Organized workshops on machine learning, deep learning, blockchain, and research methodology; coordinated collaborations with IEEE student branches and alumni to expand technical learning and networking.

Honors & Awards

Women in Tech BD, Huawei – 1st Runner-up	2024
IUTCS "The Great Web Off" – Champion	2024
Blockchain Olympiad Bangladesh – 4th, National Rank	2024
BUET CSE Fest Deep Learning Sprint 2.0 – 32nd	2024
ASR for Regional Dialects Datathon, Bengali.AI – 30th	2023
ICPC Algo Queen – Top 60, Global Rank	2023
CodeRush 1.0 IUT Programming Contest – 9th	2023
IUT Intra-University Programming Contest – 6th	2022
Ada Lovelace National Girls' Programming Contest – 17th	2021

Technical Skills

Programming: Python, C++, Java, SQL
Tools: Git, Docker, LaTeX, Jupyter, Kaggle, Matplotlib
ML/CV: PyTorch, TensorFlow/Keras, OpenCV, scikit-learn, Hugging Face, NumPy, Pandas